

The Solo Operator's AI Agent Cost Model

5 Cost Drivers. Score Your Stack Before It Scales.

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Token bills are the cost you see. Four other drivers are the ones that compound. Score your agent stack across all five — anything under 15/25 means you're leaking money at scale.

1 Model Calls — Are you on the right tier?

5 = Haiku for triage, Sonnet for reasoning, Opus reserved for complex synthesis 3 = One model for most tasks, occasionally switching 1 = Same model for everything — no tiering

The fastest cost fix: Haiku handles 60-80% of agent tasks at 1/10th the price of Sonnet.

2 Context Accumulation — How much are you sending?

5 = Summaries passed between turns; context pruned at session boundaries 3 = Partial history, some pruning 1 = Full conversation history appended each turn

A 20-turn conversation with full history can cost 20x the tokens of a single-turn call.

3 Tool Call Overhead — What's in the toolbox?

5 = Tools defined at runtime based on task; minimal base definitions 3 = Curated set, some unused tools occasionally included 1 = All tools included in every call regardless of task

10 tool definitions = 20k tokens per call before the model does any actual work.

4 Session Management — How long do agents run?

5 = Hard timeouts, completion criteria, fallback on loop detection 3 = Soft limits, usually stops when done 1 = No timeouts or loop detection — agent runs until it stops

With session-based pricing, a stuck agent at 45 minutes costs the same as 45 minutes of useful work.

5 Orchestration Layer — What's running your agents?

5 = Deterministic steps for predictable work; LLM only where judgment is needed 3 = Mix of LLM and deterministic; some unnecessary AI calls 1 = Full LLM orchestration for tasks that don't require it

The most reliable (and cheapest) parts of an agent pipeline are bash scripts and scheduled jobs.

Decision Matrix

Total Score	Recommendation
21-25	Well-optimized. Review quarterly.
15-20	Manageable. Fix before scaling.
10-14	Leaking. Audit now.
Below 10	Stop and rebuild the cost model first.

The bottleneck isn't the model. It's knowing where the money goes.